



TECHNICAL MAINTENANCE MANUAL

Fire Alarm Maintenance Manual - UH-FIRE-505

|                |                                                     |
|----------------|-----------------------------------------------------|
| Document ID    | MAN-036                                             |
| Version        | v1.0.0                                              |
| Client         | UrbanHive Facility Technologies Pvt. Ltd.           |
| Site           | GreenPark Meridian Residency, Whitefield, Bengaluru |
| Effective date | 2024-07-01                                          |
| Review cycle   | Annual or after any major incident                  |

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## 1. Equipment Overview and Technical Specifications

Addressable fire alarm control panel (FACP) serving GreenPark Meridian towers with smoke/heat detection, manual call points, hooter-strobe notification, and integration to lift recall and fire pump.

Model code: UH-FIRE-505

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 230V AC mains with 24V 7Ah battery backup 24h; loop wiring FP200 gold

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-FIRE-505, commissioning checklist CC-036, and spare parts catalogue SP-UH-FIRE-505.

Technicians must verify nameplate data matches AMIS asset record UH-FIRE-505 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

## 2. Installation Guidelines

Loop wiring in fire-rated conduit. End-of-line resistor 4.7k $\Omega$  verified per device. Cause-and-effect matrix uploaded to UrbanHive compliance portal before occupancy certificate.

### Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance  $\leq 1$  ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

## 3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: panel trouble log review; printer paper check.
- Weekly: random MCP test one per tower; verify hooter audibility at farthest apartment.
- Monthly: smoke detector sensitivity test per 10% sample using test smoke.
- Quarterly: battery discharge test 30 minutes under load.
- Annual: full cause-and-effect test with fire department witness where required.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

## 4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

#### **Fault 01 - Code F101**

Symptom: Loop short circuit trouble

Remediation: Isolate loop section; measure resistance; locate crushed cable in shaft.

#### **Fault 02 - Code F208**

Symptom: Detector head contamination alarm

Remediation: Clean chamber with approved aerosol; replace if sensitivity drift > 15%.

#### **Fault 03 - Code F315**

Symptom: MCP glass broken nuisance

Remediation: Verify tamper spring; replace element; check for vandalism pattern on floor.

#### **Fault 04 - Code F401**

Symptom: Notification appliance circuit open

Remediation: Trace strobe circuit; replace failed 24V sounder on level 12 corridor.

#### **Fault 05 - Code F509**

Symptom: Battery backup fail under load

Remediation: Replace battery bank; verify charger float 27.4V; test after 24h recharge.

#### **Fault 06 - Code F602**

Symptom: Ground fault on loop 2

Remediation: Megger loop to earth; dry junction boxes in parking level.

#### **Fault 07 - Code F701**

Symptom: Printer fault / log not printing

Remediation: Replace thermal paper; clean print head; verify ribbon sensor.

#### **Fault 08 - Code F803**

Symptom: BMS fire signal not received

Remediation: Test dry contact output; verify MODBUS mapping point 40021.

#### **Fault 09 - Code F902**

Symptom: Lift recall relay not dropping

Remediation: Test relay coil voltage; inspect fire interface wiring diagram UH-FIRE-LIFT-IF.

#### **Fault 10 - Code F1001**

Symptom: Repeated isolate events on loop 3

Remediation: Identify device address 247; replace base if intermittent short.

#### **Fault 11 - Code F1104**

Symptom: Heat detector rate-of-rise fault

Remediation: Replace detector in kitchenette refuse area; verify spacing compliance.

Fault 12 - Code F1202

Symptom: Panel firmware checksum error

Remediation: Reload approved firmware UH-FACP-2.8.3 from UrbanHive secure portal.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

| Part No. | Description              | Min stock |
|----------|--------------------------|-----------|
| SP-F01   | Smoke detector head      | 24        |
| SP-F02   | MCP break-glass element  | 30        |
| SP-F03   | 24V 7Ah battery          | 8         |
| SP-F04   | Strobe sounder combo     | 12        |
| SP-F05   | End-of-line resistor kit | 10        |
| SP-F06   | FACP printer paper roll  | 20        |

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Life safety system - no bypass without UrbanHive Fire Safety Officer written approval. Maintain detector coverage during any isolate. Report panel trouble to security within 5 minutes.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0036 (08:00-20:00 IST)

Email: support-uh-fire-505@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Fire Alarm` with asset\_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-036